Thermodynamic description of the Ni–Ti system: a synthetic benchmark document

ORCHID synthetic benchmark (generated; no human authors)

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Abstract

This synthetic document describes the Ni–Ti binary system as calculated with pycalphad from the SGTE dataset NiTi-04Tok.tdb; every figure, table and stated value is generated, and the exact values are kept in a separate truth key. The highest invariant reaction, L → (Ni) + Ni3Ti, lies at 1575.6 K.

1. Introduction

The Ni–Ti system belongs to the 20 binaries of the ORCHID benchmark, chosen to cover diagram types such as intermetallic rich, congruent compound. The thermodynamic description is the SGTE collection entry [1], which reproduces the assessment of T. Tokunaga, K. Hashima, H. Ohtani, M. Hasebe, Mater. Trans. JIM, 45, 1507-14(2004). [2]. The calculations use pycalphad 0.11.2 at 1 atm.

For Ni, the liquidus meets the axis at 1728.2 K. Pure Ti melts at 2498.8 K.

2. Phase diagram and invariant reactions

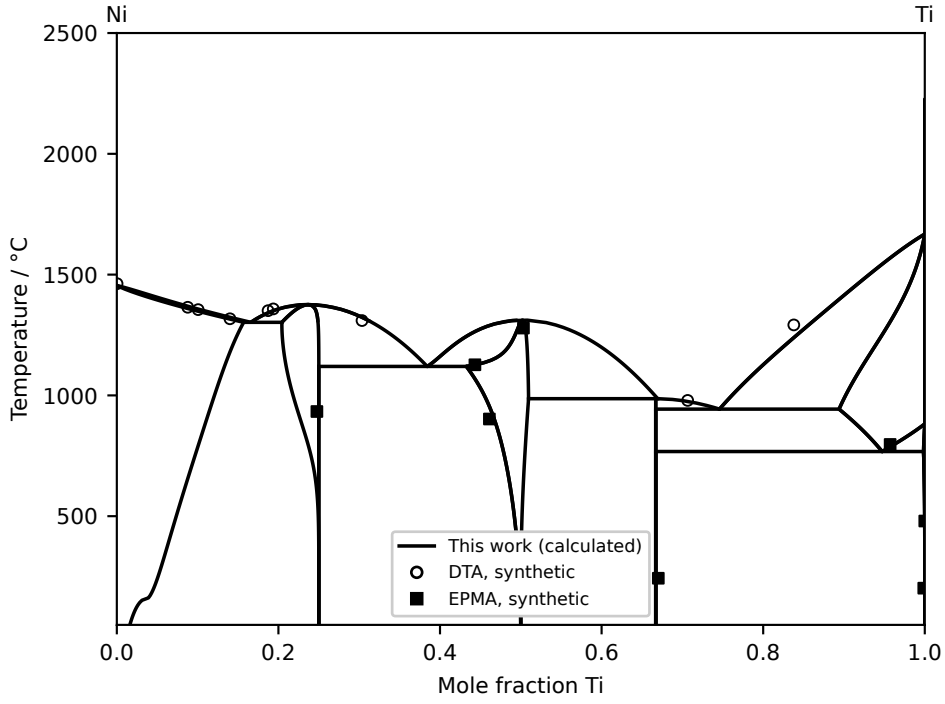


Fig. 1. Calculated Ni–Ti phase diagram with synthetic experimental-style points.

We obtain 1575.6 K for the eutectic reaction (L → (Ni) + Ni3Ti), where L has 16.6 at.% Ti. At 1393.0 K the eutectic L → Ni3Ti + NiTi takes place; the L phase then contains

38.4 at.% Ti. We obtain 1259.5 K for the peritectic reaction (NiTi + L → NiTi2), where L has 66.9 at.%. At 1216.3 K the eutectic L → NiTi2 + (βTi) takes place; the L phase then contains 74.6 at.%. We obtain 1040.8 K for the eutectoid reaction ((βTi) → NiTi2 + (αTi)), where (βTi) has 94.7 at.% Ti. Ni3Ti melts congruently at 1649.4 K and 23.6 at.% Ti. NiTi melts congruently at 1584.9 K and 50.2 at.% Ti.

Table 1 compares the calculated temperatures with experimental values from the literature [3–5]; in this synthetic document those values and references are fictitious.

Table 1. Invariant reactions and congruent points, calculated and experimental (Ni–Ti).

Reaction	Type	T / °C			Compositions (x(Ti))		
		this work	exp.	Ref.	phase 1	phase 2	phase 3
L → Ni3Ti	congruent melting	1376.2	–	–	0.236	0.236	–
L → NiTi	congruent melting	1311.8	1308.8	[3]	0.502	0.502	–
L → (Ni) + Ni3Ti	eutectic	1302.5	1298.5	[4]	0.166	0.157	0.204
L → Ni3Ti + NiTi	eutectic	1119.9	–	–	0.384	0.250	0.433
NiTi + L → NiTi2	peritectic	986.4	–	–	0.510	0.669	0.667
L → NiTi2 + (βTi)	eutectic	943.2	–	–	0.746	0.667	0.894
(βTi) → NiTi2 + (αTi)	eutectoid	767.6	760.6	[4]	0.947	0.667	0.998

Phases 1–3 in the order written in the reaction. Calculated with pycalphad 0.11.2 from the SGTE dataset; synthetic document.
Experimental temperatures are fictitious values attributed to invented references [3–5] for testing evidence-class extraction; they are not data.

3. Phase stability

Table 2. Solid phases: structure, stability range and homogeneity range (Ni–Ti).

	Structure			Stability range / °C		Homogeneity range	
Phase	Pearson	Strukturb.	Prototype	from	to	at.% Ti	at T / °C
(Ni)	cF4	A1	Cu	< 27	1455.1	0.0–15.6	1297
Ni3Ti	hP16	$D0_{24}$	Ni3Ti	< 27	1376.2	20.4–25.0	1267
NiTi	cP2	B2	CsCl	< 27	1311.8	43.3–50.9	1117

Table 2 (continued).

	Structure			Stability range / °C		Homogeneity range	
Phase	**Pearson**	**Strukturb.**	**Prototype**	**from**	**to**	**at.% Ti**	**at T / °C**
NiTi2	–	–	–	< 27	986.4	66.7	227
(βTi)	cI2	A2	W	767.6	1667.8	89.4–100.0	947
(αTi)	hP2	A3	Mg	< 27	881.8	99.8–100.0	767

Structure labels follow the TDB phase names (– where the name does not identify a structure). Stability range: temperatures between which the phase is stable in some part of the diagram.

The upper stability limit of NiTi2 is 1259.5 K. The solubility of Ti in (Ni) reaches 15.7 at.% at 1575.6 K.

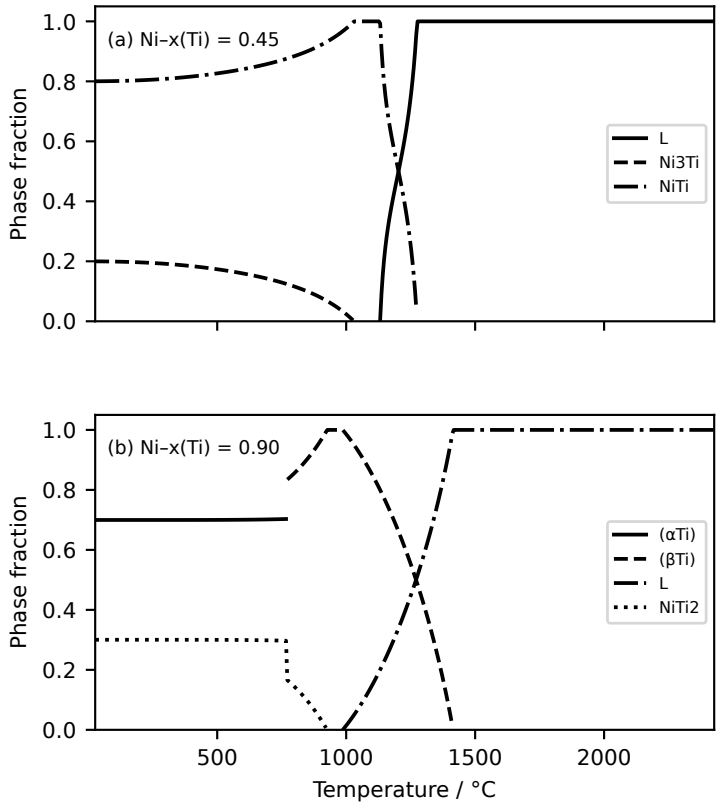


Fig. 2. Calculated equilibrium phase fractions versus temperature for Ni–x(Ti) = 0.45 and x(Ti) = 0.90.

4. Thermodynamic properties

At 2600 K the integral mixing enthalpy of the liquid has its minimum, −40840 J/mol, at 38.0 at.% Ti. The calculated enthalpy of formation of NiTi at 298.15 K is −33.34 kJ/mol per mole of atoms. Per formula unit NiTi this is −66.7 kJ/mol. Its heat capacity at room temperature is 25.47 J/(mol K). For the equiatomic liquid at 2600 K, the Gibbs energy of mixing is −30.75 kJ/mol-atom.

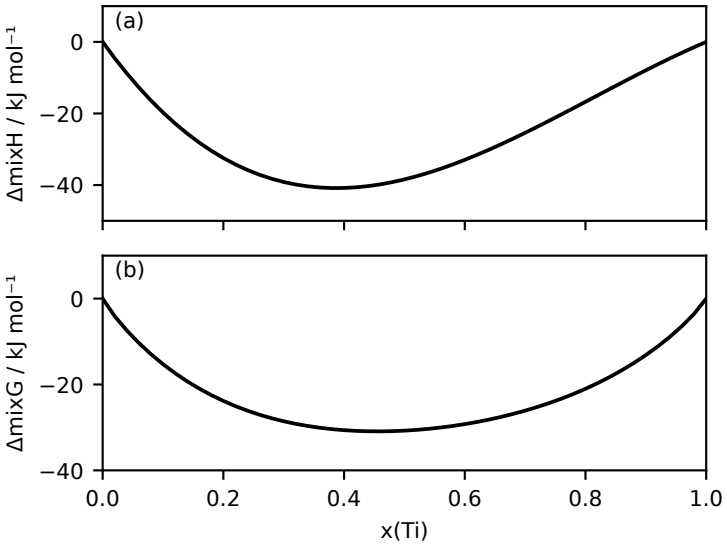


Fig. 3. Calculated (a) mixing enthalpy and (b) mixing Gibbs energy of the liquid at 2600 K.

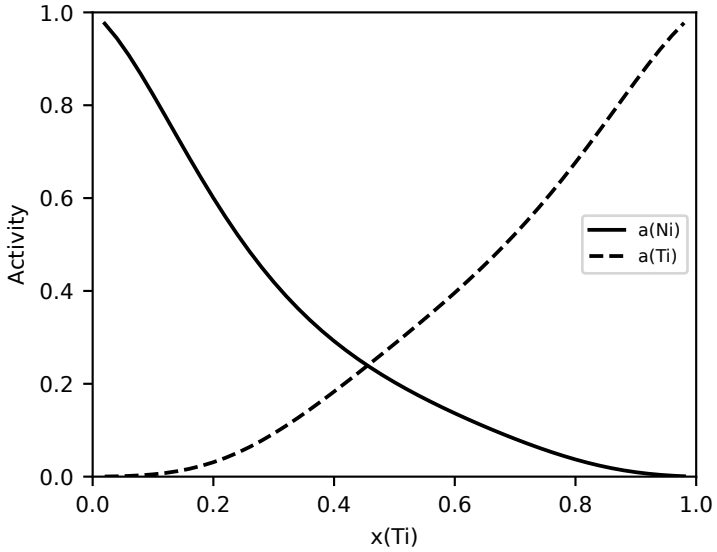


Fig. 4. Calculated activities in the liquid at 2600 K (reference: pure liquids).

Table 3. Calculated formation properties of the intermediate phases at 298.15 K (Ni–Ti).

Phase	at.% Ti	ΔfH(298.15 K) / J mol⁻¹	ΔfS / J mol⁻¹ K⁻¹	Cp(298.15 K) / J mol⁻¹ K⁻¹
Ni3Ti	25.0	−40029 ± 1000	−4.94	25.02
NiTi2	66.7	−26902 ± 1000	−1.78	25.10
NiTi	49.9	−33343 ± 1500	1.22	25.47

Table 4. Integral and partial mixing properties of the liquid at 2600 K (Ni–Ti).

	Liquid at 2600 K					
x(Ti)	**ΔmixH / kJ mol⁻¹**	**ΔmixG / kJ mol⁻¹**	**a(Ni)**	**a(Ti)**	**ΔH(Ni) / kJ mol⁻¹**	**ΔH(Ti) / kJ mol⁻¹**
0.100	−19.72	−15.27	0.8229	0.0049	−3.66	−164.27
0.300	−39.15	−28.57	0.4193	0.0928	−27.09	−67.30
0.500	−38.43	−30.75	0.2034	0.2858	−58.88	−17.97
0.700	−25.41	−26.09	0.0812	0.5228	−83.34	−0.58
0.900	−7.94	−13.15	0.0096	0.8526	−84.74	0.59

Reference states: pure liquid Ni and Ti at 2600 K; per mole of atoms. ΔH: partial molar enthalpies of mixing.

5. Conclusions

This document is synthetic and exists only to test extraction tools; its truth key lists every boundary, field, invariant, plotted point, table cell and stated value.

References

[1] B. Hallstedt, SGTE binary dataset collection, Calphad 89 (2025) 102833 (CC BY 4.0).

[2] T. Tokunaga, K. Hashima, H. Ohtani, M. Hasebe, Mater. Trans. JIM, 45, 1507-14(2004).

[3] K. Synthetica, L. Exemplar, J. Fict. Phase Equilib. 12 (1987) 345–352 — fictitious reference, invented for this synthetic benchmark.

[4] M. Placeholder, Z. Notreal, Fict. Thermochim. Acta 41 (1994) 77–89 — fictitious reference, invented for this synthetic benchmark.

[5] R. Imaginary, Proc. Synth. Calorim. Soc. 8 (2003) 1102–1110 — fictitious reference, invented for this synthetic benchmark.